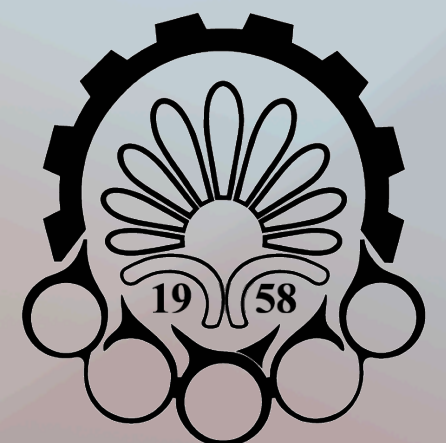


Zero-Shot Learning With CLIP

Computation Intelligence and its Application in Mechatronics

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CLIP: Connecting text and images

CLIP (Contrastive Language–Image Pre-training)

CLIP (from OpenAI) is a multimodal model trained to connect images and text in a shared embedding space. It was introduced in the paper “**Learning Transferable Visual Models From Natural Language Supervision**” (Radford et al., 2021).

Instead of training on labeled datasets like ImageNet, CLIP was trained on **400 million image–text pairs** from the internet using a **contrastive learning objective**.

Learning Transferable Visual Models From Natural Language Supervision

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Abstract

State-of-the-art computer vision systems are trained to predict a fixed set of predetermined object categories. This restricted form of supervision limits their generality and usability since additional labeled data is needed to specify any other visual concept. Learning directly from raw text about images is a promising alternative which leverages a much broader source of supervision. We demonstrate that the simple pre-training task of predicting which caption goes with which image is an efficient and scalable way to learn SOTA image representations from scratch on a dataset of 400 million (image, text) pairs collected from the internet. After pre-training, natural language is used to reference learned visual concepts (or describe new ones) enabling zero-shot transfer of the model to downstream tasks. We study

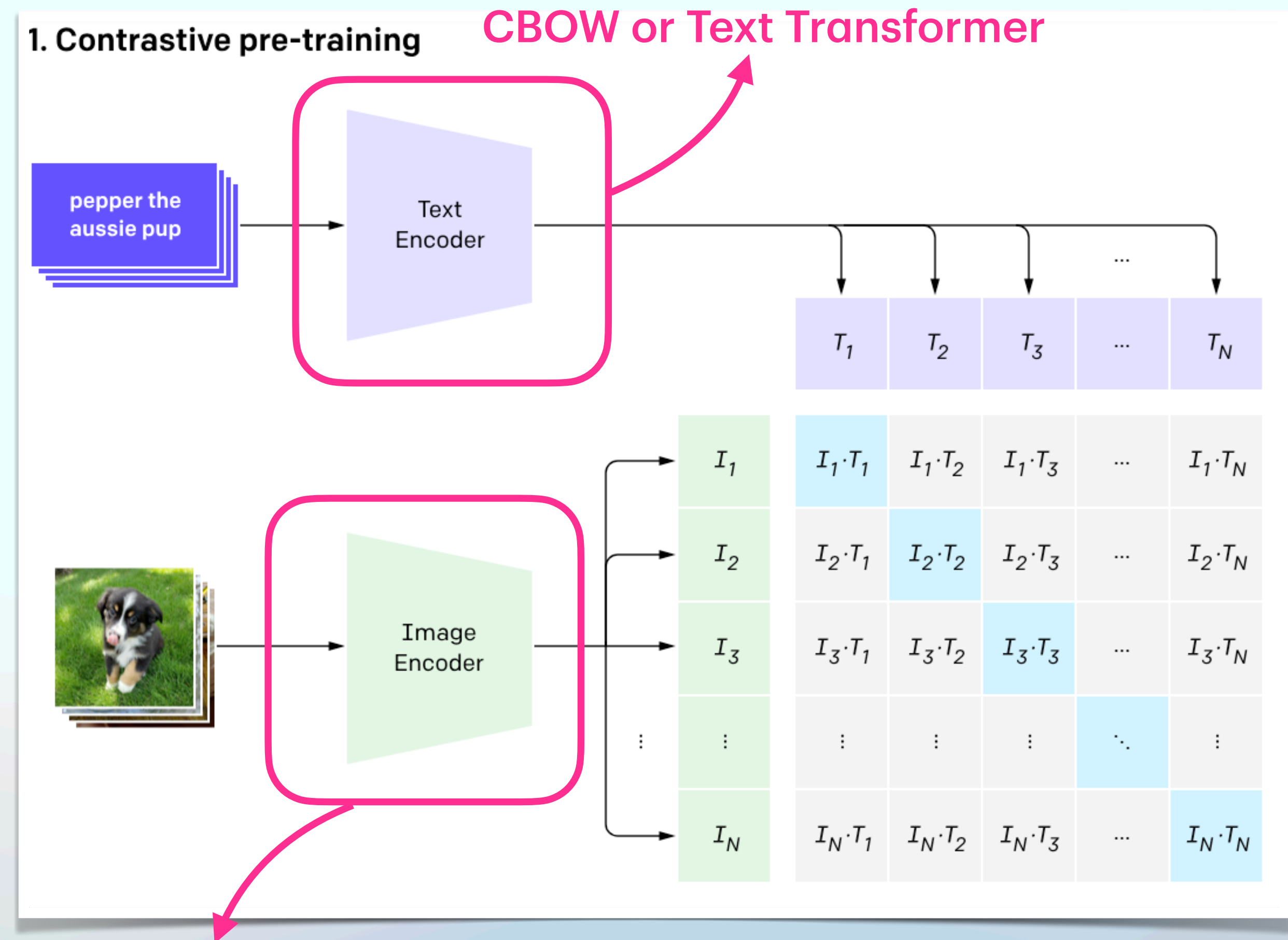
Task-agnostic objectives such as autoregressive and masked language modeling have scaled across many orders of magnitude in compute, model capacity, and data, steadily improving capabilities. The development of “text-to-text” as a standardized input-output interface (McCann et al., 2018; Radford et al., 2019; Raffel et al., 2019) has enabled task-agnostic architectures to zero-shot transfer to downstream datasets removing the need for specialized output heads or dataset specific customization. Flagship systems like GPT-3 (Brown et al., 2020) are now competitive across many tasks with bespoke models while requiring little to no dataset specific training data.

These results suggest that the aggregate supervision accessible to modern pre-training methods within web-scale collections of text surpasses that of high-quality crowd-labeled NLP datasets. However, in other fields such as computer vision it is still standard practice to pre-train models on crowd-labeled datasets such as ImageNet (Deng et al., 2009).

CLIP: Connecting text and images

How to use it?

CLIP pre-trains an image encoder and a text encoder to predict which images were paired with which texts in our dataset.



CLIP: Connecting text and images

How to use it?

We then use this behavior to turn CLIP into a zero-shot classifier. We convert all of a dataset's classes into captions such as “a photo of a dog” and predict the class of the caption CLIP estimates best pairs with a given image.

1. Contrastive pre-training

